

Gemini Chat

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [150, 152, 146, 163, 163, 161, 154, 167, 156, 171]

Tap amplitudes (pixels): [127.94139283281233, 122.80065146407001, 126.21014222319853, 94.5568612000208, 111.07204868912791, 117.6860229593982, 95.1892851112981, 88.3911760301898, 94.57801012920498, 102.45974819410792, 78.39005038906404]



Based on the extracted features provided, here is the analysis and scoring according to the MDS-UPDRS Item 3.4 criteria.

Analysis of Features

1. Rhythm and Speed (Tap Periods):

- The tap periods show a general trend of lengthening (slowing down) over the duration of the test.
- The sequence begins around **150 frames** and ends around **171 frames**.
- While there is variation (e.g., speeding up slightly at the 7th interval), the overall trend indicates **mild slowing**.
- There are no extreme outliers in the period data (e.g., a period 2-3x longer than the average), suggesting there were no distinct "halts" or "freezes."

2. Amplitude (Tap Amplitudes):

- **Taps 1–3:** The amplitude is relatively stable and high (approx. 122–128 pixels).

- **Tap 4:** There is a significant drop in amplitude (from ~126 down to ~94 pixels). This occurs early in the sequence but not immediately after the first tap.
- **Taps 5–11:** The subject attempts to recover amplitude (reaching ~117 at Tap 6) but generally sustains a lower average amplitude than the start, finishing at the lowest point (78 pixels) at the very end.
- The distinct drop occurs around the 4th tap, which can be interpreted as **decrementing midway** through the sequence, rather than just at the very end.

Scoring Rationale

- **Not Score 0 (Normal):** There is evidence of slowing and amplitude reduction.
- **Not Score 1 (Slight):** The amplitude drop occurs at Tap 4, which is earlier than "near the end" of the 10-tap sequence.
- **Matches Score 2 (Mild):** The data fits two criteria for a Mild rating:
 - **Mild slowing:** The period length increases from ~150 to ~171.
 - **Amplitude decrements midway:** The substantial drop in amplitude happens around the 4th tap, followed by irregular, generally lower amplitudes.
- **Not Score 3 (Moderate):** The amplitude does not decrement immediately after the 1st tap (the first three taps are stable), and there are no long arrests/freezes in the period data.

Score: 2

